TEMASEK JUNIOR COLLEGE

PRELIMINARY EXAMINATION

Higher 2

GEOGRAPHY

9173/02

Paper 2

18 September 2025

Additional Materials: Insert

3 hours

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Section A.
Answer **Question 1**.

Section B
Answer **Question 2**.

Section C
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **4** printed pages

Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4

Section A

Cluster 4 Fieldwork

1 A group of 24 students chose to investigate flood risk along the Hawkcombe stream in Somerset, a rural area in the United Kingdom.

They proposed the following hypothesis:

"Flood risk increases downstream along the Hawkcombe stream."

The students collected data at six sites along the stream on a weekday afternoon in September. They travelled between the sites in minibuses and each team of four students disembarked to walk to the site. They took photos of the site for visual analysis and started taking measurements as soon as they could in order not to waste time.

They measured channel width and depth to calculate the wetted perimeter. To measure the river's wetted perimeter, the teams used the following equipment:

- Tape measure
- Metre rulers

The teams laid the unweighted tape measure along the riverbed and took one depth measurement in the middle of the river. This data was used to calculate the river's wetted perimeter. Their results are shown in Resource 4.

After conducting the river fieldwork, the students conducted a questionnaire survey with residents about flood risk perceptions of the Hawkcombe stream near their sampled site.

Resource 1 shows the flood risk map of the Hawkcombe stream in Somerset. Resource 2A shows Site 1. Resource 2B shows Site 6. Resource 3 shows the questionnaire survey. Resource 4 shows the results of channel characteristics measured at six sites.

(a) Explain why the hypothesis chosen by the students is appropriate for investigating flood risk along the Hawkcombe stream. [4]

(b) With reference to Resources 2A and 2B, suggest two potential risks to students' personal safety when collecting data at both sites, and explain how these risks could be minimised. [6]

(c) Using Resources 2A and 2B, suggest how physical characteristics of the channel and its location could influence flood risk at Sites 1 and 6. [4]

(d) Using Resource 3, explain the strengths and limitations of the questionnaire survey in providing data about flood risk perceptions of the Hawkcombe stream. [6]

(e) Use a suitable data technique to represent the changes in wetted perimeter downstream along the Hawkcombe stream shown in Resource 4. [5]

(f) In addition to the data collected by the students, suggest other types of data that they could have collected to help them in their investigation of flood risk. [5]

(g) The students have concluded that Site 6 has the highest risk of flooding. Using all the resources provided, evaluate the validity of this conclusion. [10]

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 5 shows the number of cities in Africa with a population of over 1 million in 1990 and 2018. Resource 6 shows the municipal solid waste destinations for selected macro-regions in 2020. Resource 7 shows a cartoon on the effects of climate change. Resource 8 shows Bosco Verticale, vertical forest apartment and buildings in Milan, Italy.

(a) Using Resource 5, describe the changes in the distribution of cities in Africa with a population of over 1 million in 1990 and 2018. [4]

(b) Compare the municipal solid waste destinations among the macro-regions in 2020 as shown in Resource 6. [5]

(c) Explain how recycling could affect the demand placed on natural environments. [5]

(d) With reference to Resource 7, explain why the impacts of climate change might differ across countries. [5]

(e) Using examples, explain the economic reasons for urban reimaging efforts. [5]

(f) Using Resource 8, explain how the urban development shown serves as both a mitigation and adaptation strategy in response to climate change. [6]

Section C

Answer **EITHER** question 3 **OR** question 4.

Cluster 3 Sustainable Future and Climate Change

3 'Social issues pose the greatest challenge to urban liveability.'

Evaluate this statement. [20]

4 Evaluate the relative effectiveness of the key actors involved in climate change responses. [20]

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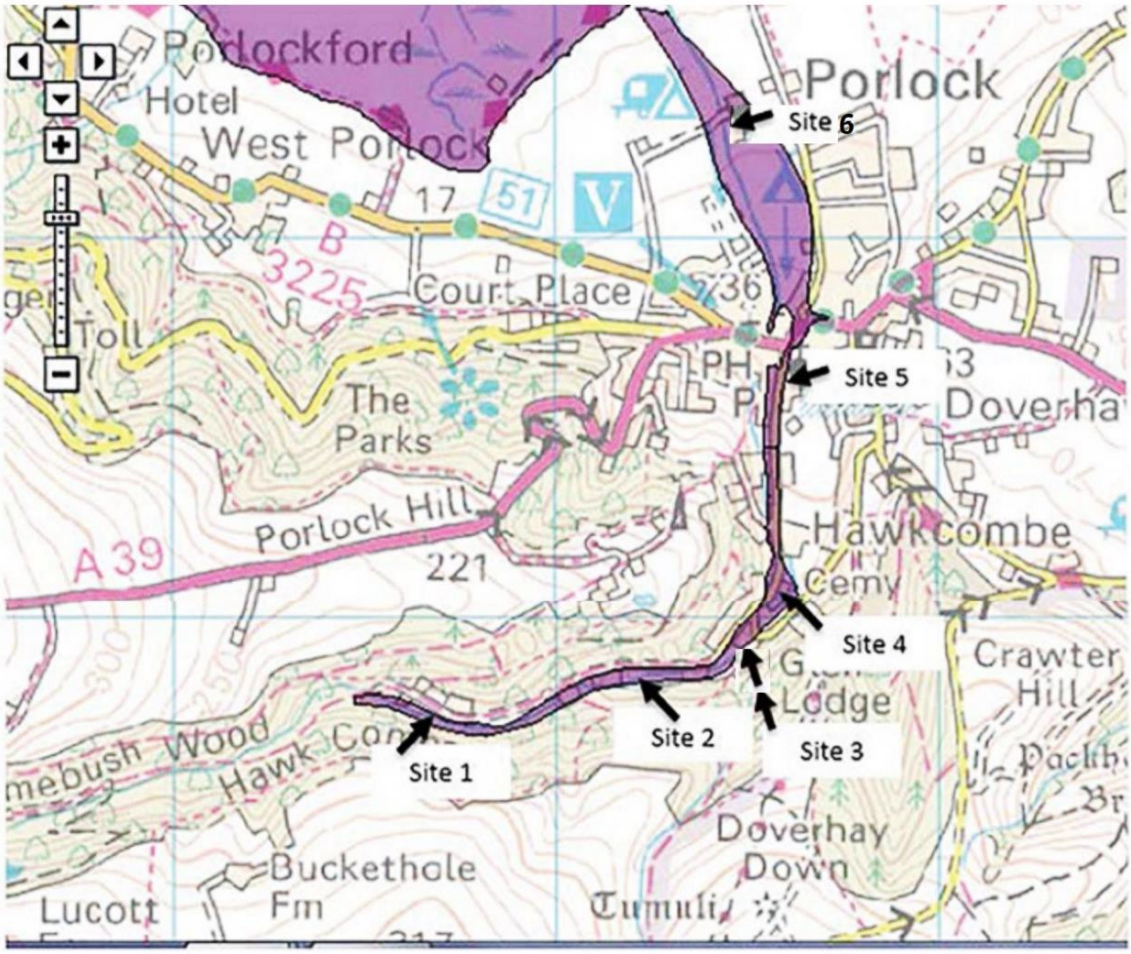
3 hours

READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

Resource 1 for Question 1

Flood risk map of the Hawkcombe stream* in Somerset



Site 1 is in the upper segment of Hawkcombe stream while Site 6 is in the lower segment of the Hawkcombe stream.

Resource 2A for Question 1
Site 1



Resource 2B for Question 1
Site 6



Resource 3 for Question 1

Questionnaire Survey

Place of residence:_____ **Age: _____**

1. How frequent do you visit Hawkcombe River?
☐ Frequently (weekly or more)
☐ Occasionally (monthly)
☐ Rarely (a few times a year)

2. Where do you currently reside?
☐ Within 1 km of the Hawkcombe River
☐ Within 1–5 km of the Hawkcombe River
☐ More than 5 km away from the Hawkcombe River

3. How likely do you think it is that a flood will occur along the Hawkcombe River within the next year?
1 ☐ Very likely
2 ☐ Likely
3 ☐ Neutral
4 ☐ Unlikely
5 ☐ Very unlikely

4. How concerned are you about the potential impact of flooding from the Hawkcombe River on your property and daily life?
1 ☐ Extremely concerned
2 ☐ Very concerned
3 ☐ Moderately concerned
4 ☐ Slightly concerned
5 ☐ Not concerned at all

5. Are you aware of the existing flood control projects and naturalized river designs implemented along the Hawkcombe River?
1 ☐ Extremely aware
2 ☐ Very aware
3 ☐ Moderately aware
4 ☐ Slightly aware
5 ☐ Not aware

6. How much trust do you have in the effectiveness of current flood management infrastructure along the Hawkcombe River to prevent significant flooding?
1 ☐ Complete trust
2 ☐ Trust
3 ☐ Neutral
4 ☐ Little trust
5 ☐ No trust

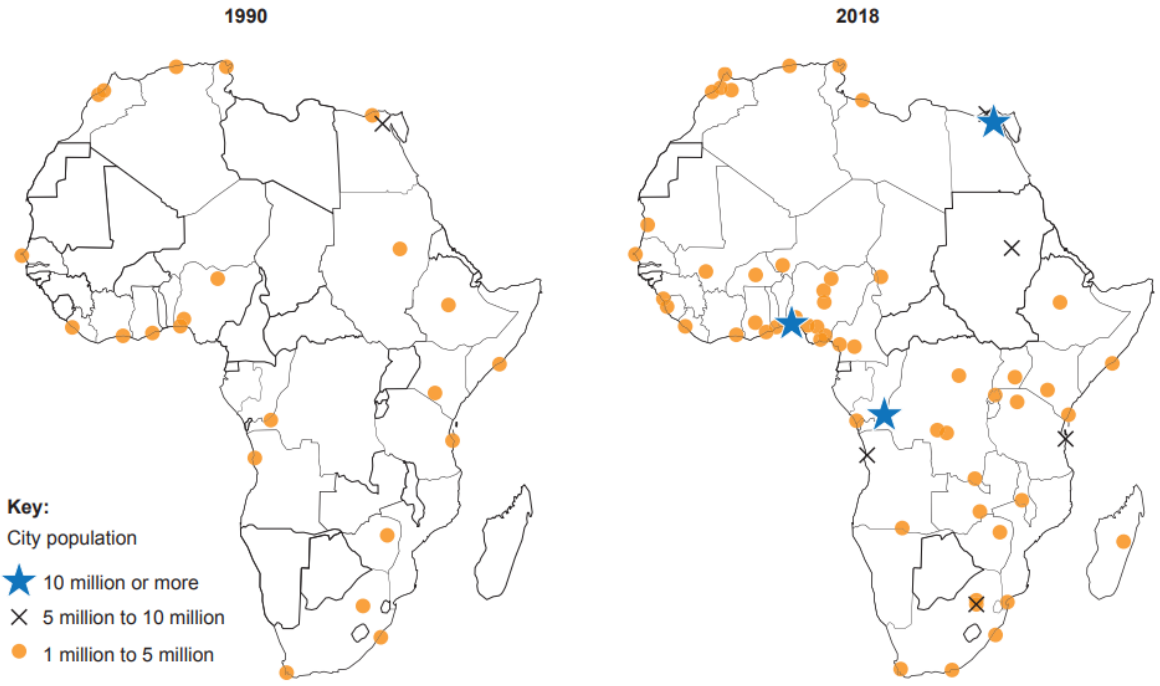
Resource 4 for Question 1

Collated Data from the six sites

Site	Width (metres)	Depth (metres)	Wetted perimeter = W + 2D (metres)
1	2.2	0.07	2.34
2	3.1	0.28	3.66
3	2.1	0.40	2.90
4	2.2	0.47	3.14
5	2.5	0.35	3.20
6	4.8	2.56	9.92

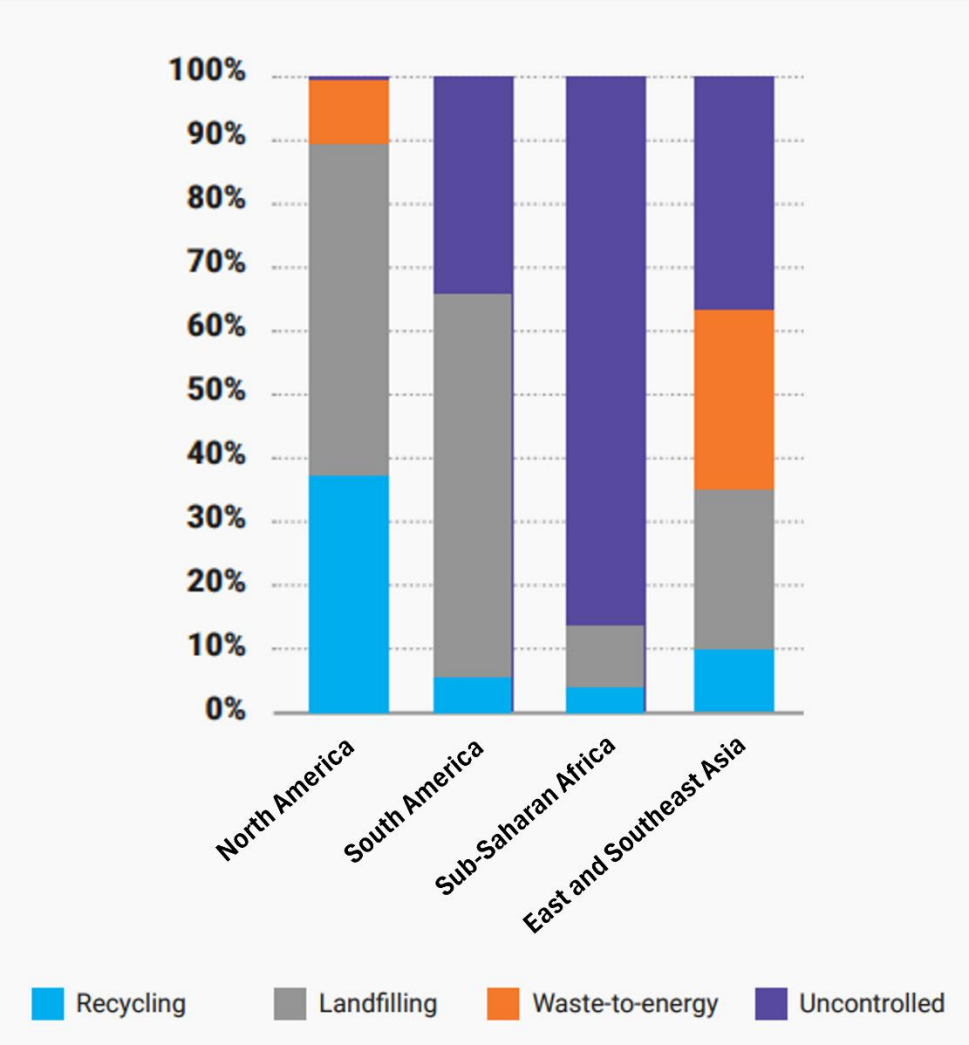
Resource 5 for Question 2

Cities in Africa with a population over 1 million, 1990 and 2018



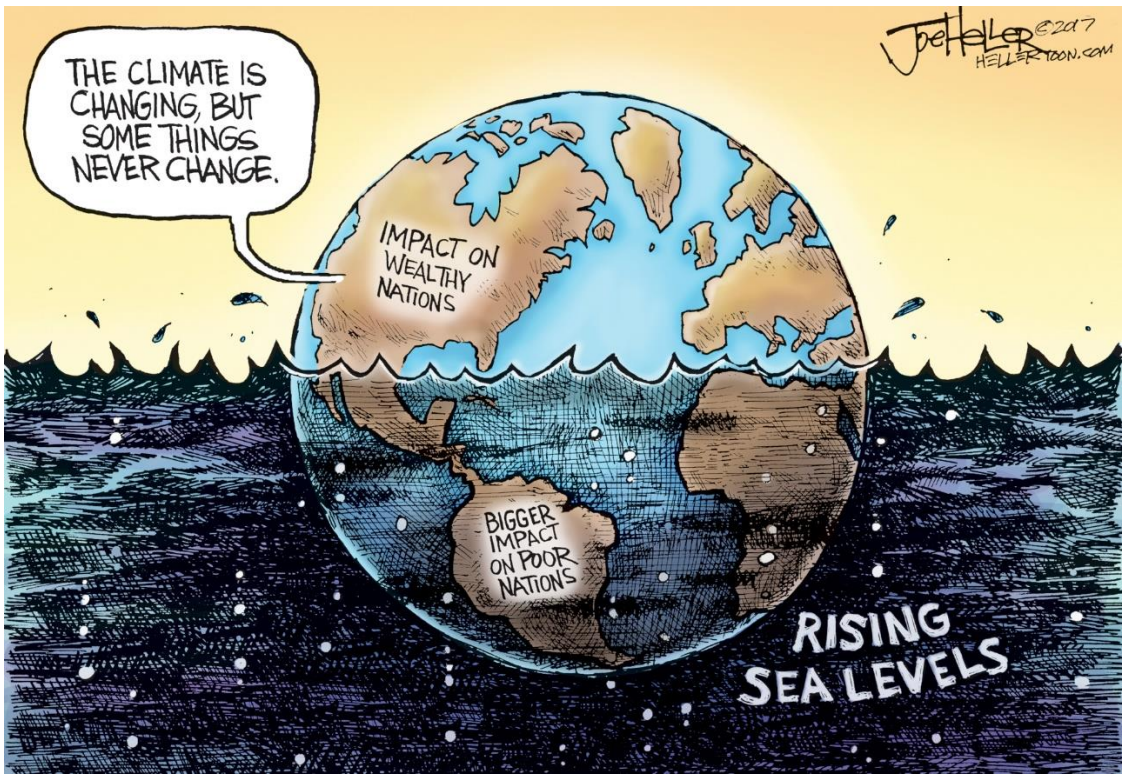
Resource 6 for Question 2

Municipal solid waste destinations for selected macro-regions, 2020



Resource 7 for Question 2

Cartoon on effects of Climate Change



Resource 8 for Question 2

Bosco Verticale, vertical forest apartment and buildings in Milan, Italy





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